



HACK FUSION

— ONE CHALLENGE. COUNTLESS SOLUTIONS. —

**Project Title:
Digital Citizen Assistant
For Multilingual Access To
Government Services And
Schemes**

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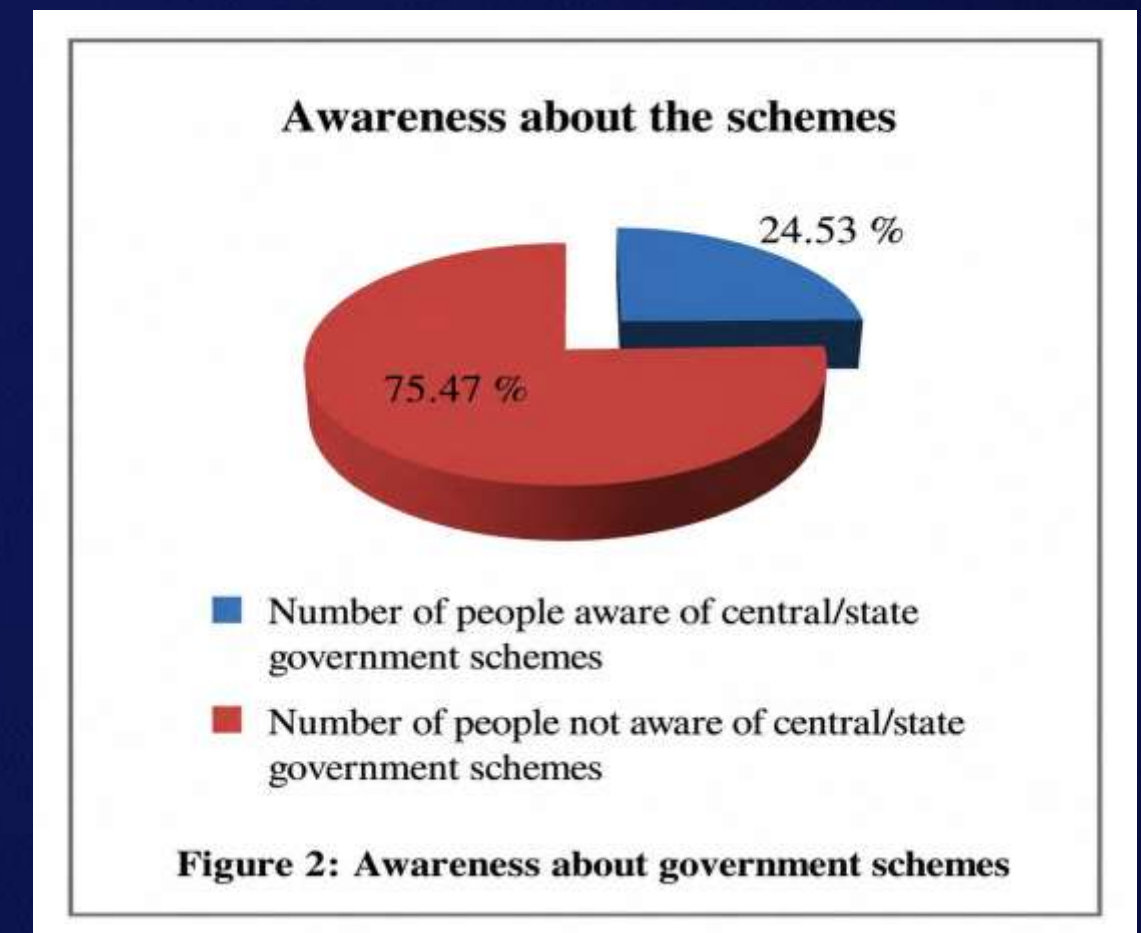
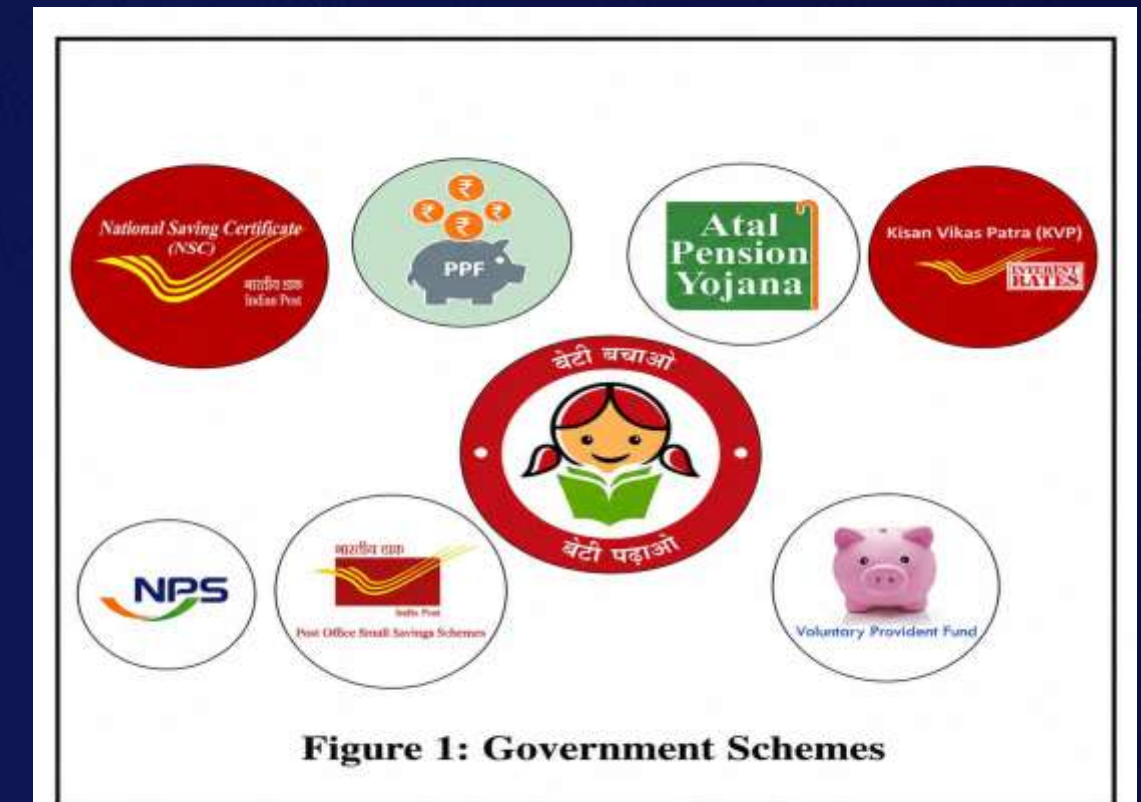
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Problem Statement

Over 800 million Indians are eligible for government welfare schemes, but millions never receive these benefits. With 3,000+ central, state, and district level schemes in existence, there is **no unified platform** to help citizens.

Core Gaps in the Current System:

- ❖ **No Automation:** No automation in scheme discovery, eligibility verification, or application guidance. Everything is manual.
- ❖ **No Accessibility Support:** No support for illiterate users, elderly citizens, rural people or people with disabilities.
- ❖ **Complexity & Confusion:** Complex scheme information and application procedures cause user confusion.
- ❖ **No Proactive Alerts:** No proactive alerts for application deadlines, renewals, or new scheme updates.
- ❖ **No Personalization:** No personalized recommendations, benefit score, or AI prediction for missed schemes.
- ❖ **Language & Tracking Barriers:** Language barriers and lack of real-time application status tracking limit citizen access.



Proposed Solution:

SUGAM SEVA:

“Easy Access to Government Services”

Sugam Seva is the **Digital Citizen Assistant** which is a **multimodal AI platform** designed to make government welfare schemes accessible to every citizen, offering **two service modes** within a single system.

Citizen App (Mobile Application)

- ❖ Intended for citizens comfortable using smartphones and digital services.
- ❖ Users securely connect **DigiLocker** or upload documents.
- ❖ AI identifies eligible schemes automatically using documents and User's Profile.
- ❖ AI simplifies complex information into easy to understand language.
- ❖ AI automatically prepares applications.
- ❖ **AI voice agent** explains schemes, answers queries, and collects approval before proceeding.
- ❖ Final submission remains under the **citizen's control**.

Government Dashboard

- ❖ Designed for citizens needing assistance at **Government Offices** or **Common Service Centres (CSCs)**.
- ❖ Government officials authenticate using **biometrics**.
- ❖ Officials help citizens with registration
- ❖ Documents remain securely stored in **DigiLocker**.
- ❖ Documents are accessible **only with citizen's consent**.
- ❖ **AI voice calling engine** automatically calls to citizens when scheme approaches.
- ❖ Ensures citizens receive support **without losing control** over their application.

Features

AI Powered Scheme Matching:

Automatically analyzes citizen profiles and documents to identify eligible government schemes from trusted sources like MyScheme, UMANG, and state portals.

Personalized Benefits Score:

AI generates a score that predicts missed benefits and ranks the most relevant schemes for each citizen.

Multilingual AI Voice Assistant:

Explains schemes, answers citizen queries, and guides users through the entire process in their preferred regional language.

Secure DigiLocker and OCR Integration:

Citizens link DigiLocker or upload documents via OCR. Documents stay securely with the citizen and never retained on local servers.

Biometric Authentication:

Secures identity verification for citizens and government officials at every sensitive step such as login, document access, and application closure.

Smart Alerts and Real Time Application Tracking:

Notifies citizens of eligibility, deadlines, and renewals, while providing live status updates from application to approval.

Idea/Approach:

Tech Stack:

- **Frontend:** React Native + Expo, React + TypeScript + Vite, Tailwind CSS, shadcn/ui, TanStack Query, Zustand, Recharts
- **Backend:** Python + FastAPI, Keycloak, Celery
- **Database:** PostgreSQL, pgvector, Redis
- **AI & ML:** Claude LLM, Embeddings, RAG, Rule Engine, AI Ranking, PaddleOCR
- **Government Integration:** DigiLocker, APISetu, Bhashini, Biometric Integration, MyScheme, UMANG, Seva Sindhu, MahaDBT, MeeSeva, District/Local Government Portals
- **Notifications:** SMS Gateway, Firebase FCM
- **Deployment:** Docker, NIC MeghRaj / Government Cloud

APIs / Libraries:

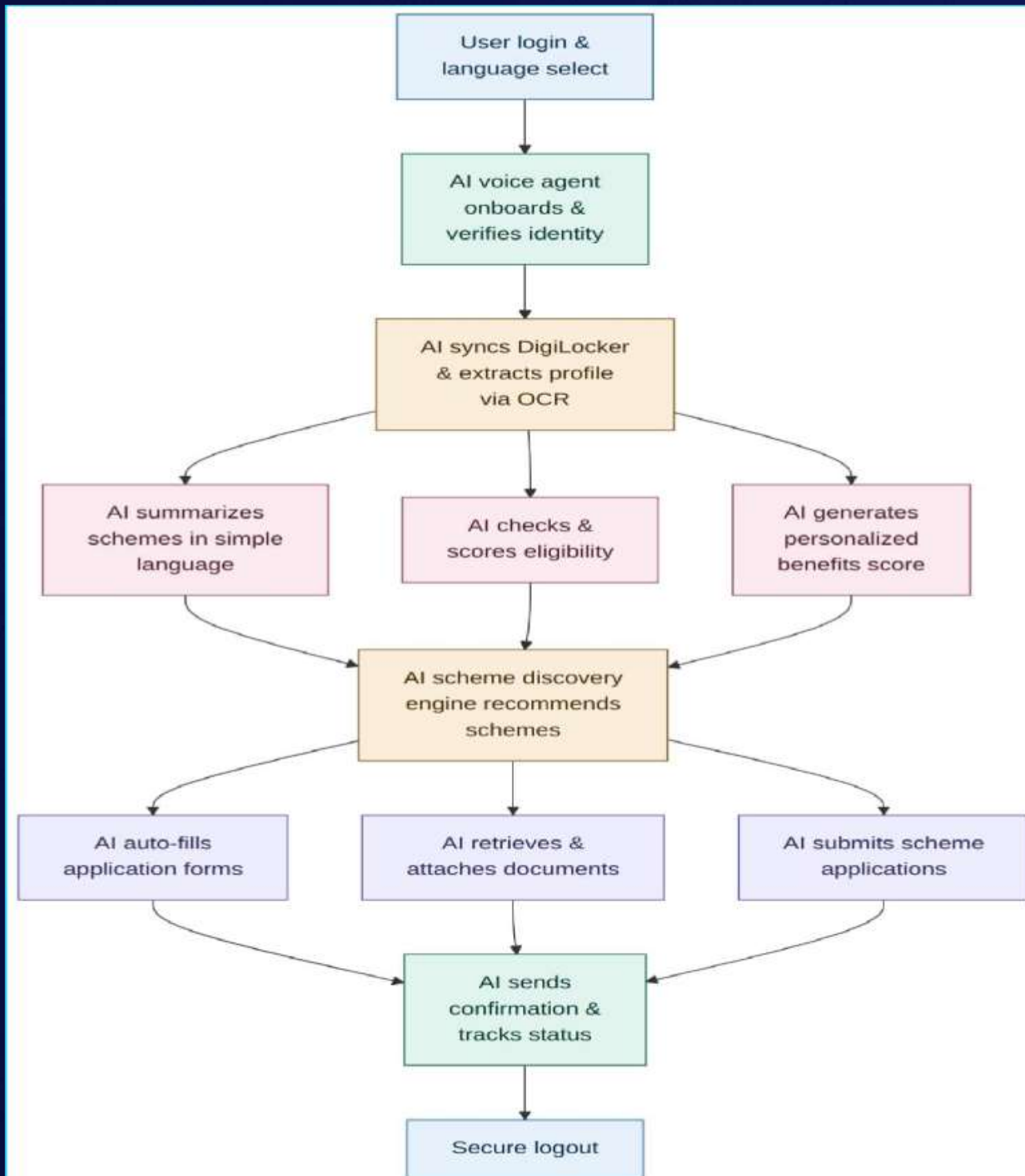
- **DigiLocker API + APISetu :** Verified citizen document access
- **Bhashini APIs :** Translation, Speech-to-Text and Text-to-Speech
- **Claude LLM API :** AI assistance and scheme summarization
- **Biometric API :** Secure identity verification
- **SMS Gateway API :** Application and scheme alerts
- **Firebase FCM API :** Push notifications
- **PaddleOCR Library :** Document information extraction
- Recharts Library : Dashboard analytics

Deployment Plan:

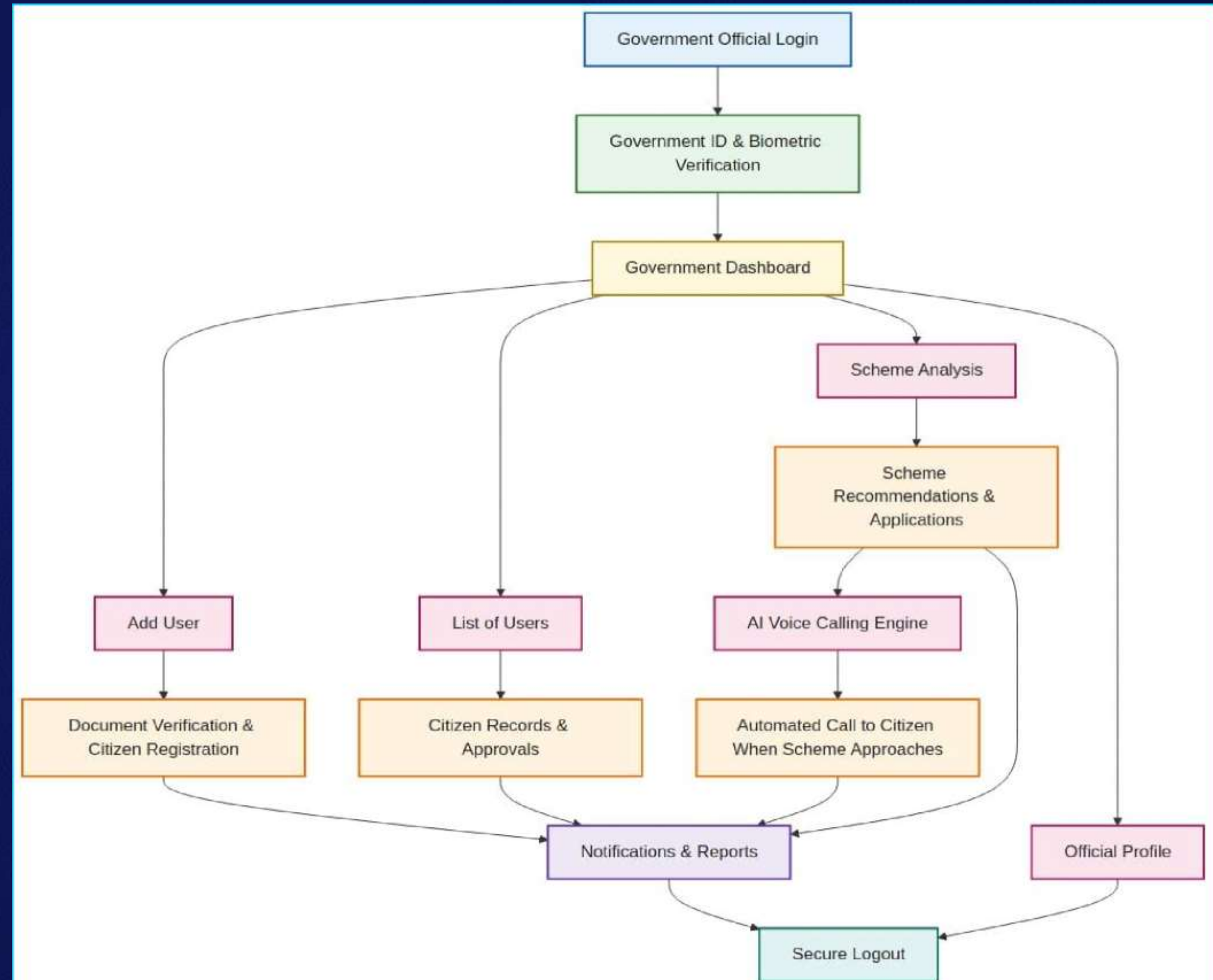
- ❑ **Citizen Mobile App**
 - Build the app using React Native + Expo.
 - Create the production Android build.
 - Publish through the Play Store / authorized distribution.
 - Connect it to the FastAPI backend.
- ❑ **Government Dashboard**
 - Build the dashboard using React + TypeScript + Vite.
 - Create the production web build.
 - Deploy it on NIC MeghRaj / Government Cloud.
 - Provide secure access to authorized officials.
- ❑ **Backend & Services**
 - Containerize FastAPI, Keycloak and Celery using Docker.
 - Deploy them on the Government Cloud.
 - Deploy PostgreSQL, pgvector and Redis.
 - Connect AI, Government and Notification services.

System Architecture / Workflow

Citizen Mobile App Workflow



Government Dashboard Workflow



Prototype:

Github Link:

Citizen App: https://github.com/chatuphotos123/SugamSeva_CitizenApp

Government Dashboard: https://github.com/chatuphotos123/SugamSeva_Govt



Citizen App

Your Benefits Score

50%

Eligible for 3 of 6 available schemes

Schemes You're Eligible For

AI-summarized in plain language — click a scheme to read more or apply.

PM Kisan Samman Nidhi

Income support of Rs.6000 per year for small and marginal farmer families.
Max income: ₹2,50,000 • Age: 18-120 • Category: all • State: all

Proceed

Disability Pension Scheme

Monthly financial assistance for citizens with certified disabilities.
Max income: ₹3,00,000 • Age: 18-120 • Category: all • State: all

Proceed

All Available Schemes

PM Kisan Samman Nidhi

Income support of Rs.6000 per year for small and marginal farmer families.
Max income: ₹2,50,000 • Age: 18-120 • Category: all • State: all

National Old Age Pension Scheme

Monthly pension support for citizens above 60 years living below the poverty line.
Max income: ₹1,00,000 • Age: 60-120 • Category: all • State: all

Scholarship for SC/ST Students

Financial assistance for SC/ST students pursuing higher education.
Max income: ₹2,00,000 • Age: 15-30 • Category: sc • State: all

Disability Pension Scheme

Monthly financial assistance for citizens with certified disabilities.
Max income: ₹3,00,000 • Age: 18-120 • Category: all • State: all

MahaDBT Girl Child Education Scheme

Support for girl child education in Maharashtra state.
Max income: ₹1,50,000 • Age: 6-18 • Category: all • State: maharashtra

Government Dashboard

1

Citizens Enrolled

46

Problems Solved

2

Active Grievances

Dashboard Home

+ Add User

List of Users

Vote of Problems

Schemes Near You

Official Profile

Top Community Problems

Infrastructure / Water Supply Grievance

42 votes

Street Lighting Not Working - Ward 12

27 votes

Delay in Ration Card Issuance

18 votes

1

Citizens Enrolled

46

Problems Solved

2

Active Grievances

Dashboard Home

+ Add User

List of Users

Vote of Problems

Schemes Near You

Official Profile

Vote of Problems

Search problems...

Add a new problem...

#	PROBLEM	VOTES	SOLVED	
1	Infrastructure / Water Supply Grievance	42	15	Vote
2	Street Lighting Not Working - Ward 12	27	27	Vote
3	Delay in Ration Card Issuance	18	4	Vote

Dashboard Home

+ Add User

List of Users

Vote of Problems

Schemes Near You

Official Profile

Add User

Name

Age

Phone

Annual Income (Rs.)

Address

General

No Disability

State (e.g. Karnataka)

Document Cards

Aadhaar

PAN

Passport

Fingerprint not yet captured

Capture Fingerprint

1

Citizens Enrolled

46

Problems Solved

2

Active Grievances

Dashboard Home

+ Add User

List of Users

Vote of Problems

Schemes Near You

Official Profile

Schemes Analyzer

SCHEME	ELIGIBLE & UNAPPLIED	USED	MISSED
PM Kisan Samman Nidhi	1	0	1
National Old Age Pension Scheme	0	0	0
Scholarship for SC/ST Students	0	0	0
Disability Pension Scheme	1	0	1
MahaDBT Girl Child Education Scheme	0	0	0
Seva Sindhu Farmer Support (Karnataka)	1	0	1

Innovation & Existing Solutions

Point	Existing Portals (MyScheme, UMANG, Seva Sindhu, MahaDBT, MeeSeva)	Our Solution (Sugam Seva)
Scheme Discovery	Citizens manually search multiple portals, with no single view of their eligibility.	AI checks the citizen profile across schemes and shows only eligible benefits.
Accessibility	Users must read and navigate forms independently, excluding many elderly, illiterate and disabled citizens.	Multilingual AI voice guidance and an Assisted Portal provide step-by-step support.
Scheme Information	Dense policy documents make scheme benefits and eligibility difficult to understand.	AI converts policy documents into short, plain-language explanations.
Alerts & Reminders	Citizens may miss deadlines, renewals and newly launched schemes without proactive alerts.	Smart SMS and push notifications warn citizens about eligibility, deadlines and renewals.
Personalization	Generic scheme lists do not show which benefits a particular citizen may be missing.	AI creates a personalized Benefits Score and predicts unclaimed eligible benefits.
Application Tracking	Limited status visibility and language barriers make applications difficult after submission.	DigiLocker/OCR documents, AI form filling and multilingual real-time tracking simplify applications.

Impact & Future Scope

Expected Impact:

- **Reduced Missed Benefits:** Citizens discover schemes they didn't know they qualified for.
- **Inclusive Reach:** Rural, elderly, and disabled citizens gain access to the system.
- **Faster Discovery:** Eligible schemes are matched automatically, not searched manually.
- **Simplified Understanding:** Scheme details are explained in plain language.
- **Transparent Tracking:** Citizens see their application status in real time.
- **Administrative Visibility:** Officials get data to guide outreach efforts.

Scalability:

- **Cloud-Native Backend:** The system scales as citizen load increases.
- **Async Processing:** Background tasks keep the app responsive under load.
- **Growing Scheme Corpus:** New schemes are added without redesigning the system.
- **Fast Response at Scale:** Caching keeps performance stable as users grow.
- **Modular State Integrations:** New states are onboarded independently.
- **Dual-Mode Growth:** The app and Assisted Portal scale through separate channels.

Future Enhancements:

- **Blockchain Verification:** Approvals and disbursements are recorded on a tamper-proof ledger.
- **Offline Support:** SMS and USSD flows reach citizens with poor connectivity.
- **Policy Feedback Loop:** Eligibility data helps policymakers improve scheme design.
- **Enhanced Accessibility:** Screen-reader and sign-language support are added.
- **Cross-Scheme Bundling:** Related schemes are recommended together.
- **Predictive Eligibility:** Future scheme matches are forecast ahead of need.

Thank You